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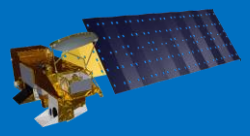
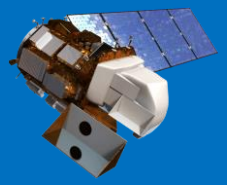
Earth Observations



Aqua MODIS
2002-2014



Landsat 8
2013-2014

Photo Credit: Dr. Jennifer L. Graham

Moving Forward

Supplementing environmental monitoring efforts with analyses of Earth observations is especially important in regions where little *in-situ* data exists. Free and open access data, like Landsat and MODIS imagery, allow for improved environmental monitoring. Many research initiatives in the southern Gulf of Mexico have limited terms of use for their datasets. If we can foster a scientific community that freely shares information, we would be able to improve environmental modeling and decision-making at the state level. This initiative would benefit the larger community by helping reduce the harmful health and economic effects of hypoxic events and algal blooms.

Photo Credit: Nara Souza

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DEVELOP National Program

Applied Sciences' Capacity Building

Gulf of Mexico Water Resources

Utilizing NASA Earth Observations to detect factors contributing to hypoxic events in the southern Gulf of Mexico

http://develop.larc.nasa.gov

DEVELOP

The Applied Science's DEVELOP National Program addresses environmental and policy issues through interdisciplinary research projects that apply NASA Earth Observations to community concerns around the globe. DEVELOP bridges the gap between NASA Earth Science and society, building capacity in both its interns and partner organizations to better prepare them to handle the challenges that face our society, along with future generations.

Partners

Consorcio de Instituciones de Investigación Marina del Golfo de México y del Caribe (CiiMar-GoMC)
Centro del Cambio Global y la Sustentabilidad en el Sureste (CCGSS)
Comisión Nacional para el Conocimiento y Uso de la Biodiversidad (CONABIO)
Centro Nacional de Datos Oceanográficos (CENDO)
Universidad Autónoma de Baja California (UABC)
Universidad Juárez Autónoma De Tabasco (UJAT)
Secretaría de Marina (SEMAR)

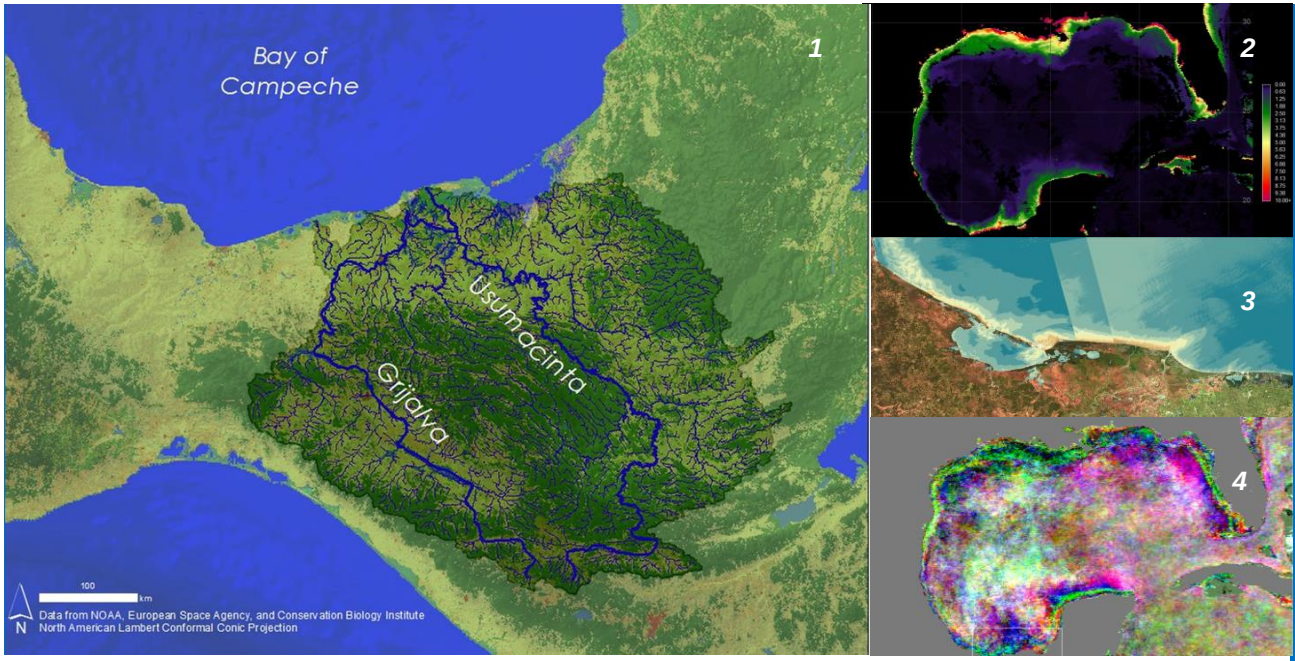
Study Areas

Analyses were performed in three separate study areas: the Grijalva-Usumacinta River Basin, the southern coast of the Bay of Campeche, and the Gulf of Mexico as a whole.

Objectives

Project objectives:
(1) Use NASA Earth observations to identify indicators of hypoxic events and algal blooms in the southern Gulf of Mexico.
(2) Model nutrient and sediment loading from the Grijalva-Usumacinta River Basin.

1. Image of the Grijalva-Usumacinta Watershed 2. MODIS derived Chlorophyll a image for November 9-16th, 2009 3. Turbidity index applied to Landsat 8 imagery 4. Amplitude image from seasonal trend analysis of CDOM



Concerns

Dead zones, hypoxic events, and harmful algal blooms in the southern Gulf of Mexico affect the health and economic well being of millions of people who live in nearby coastal communities. These events have been implicated in eutrophication, fish kills, economic loss for fisheries, and human illness. Increased nutrient and sediment influx from nearby watersheds are known to contribute to the timing and occurrences of algal blooms. However, these values have not been quantified for the Grijalva-Usumacinta River Basin. Thus, the main drivers leading to the increasing frequencies of hypoxia in this region have yet to be identified.

Decision Support

The tools used in this project will be beneficial for local agencies and partner organizations to mitigate the impacts of future algal blooms and hypoxic events in the southern Gulf of Mexico. First, inter-annual and seasonal trend analyses of algal bloom indicators could potentially be used to forecast the timing and occurrences of these events. Second, the turbidity index could continually be applied to Landsat 8 imagery as a method of remote water quality monitoring in the region. Finally, nutrient and sediment outputs from the Soil Water Assessment Tool (SWAT) could be used to revise local water quality standards.